

ECMO Configurations

Let's now continue our exploration of the fundamentals of extracorporeal membrane oxygenation (ECMO) support by discussing cannulation configurations. To remind ourselves, the basic function of the ECMO circuit is to drain deoxygenated blood, remove CO_2 /oxygenate in the membrane oxygenator, and return that blood back to the body (Fig. 8.1).

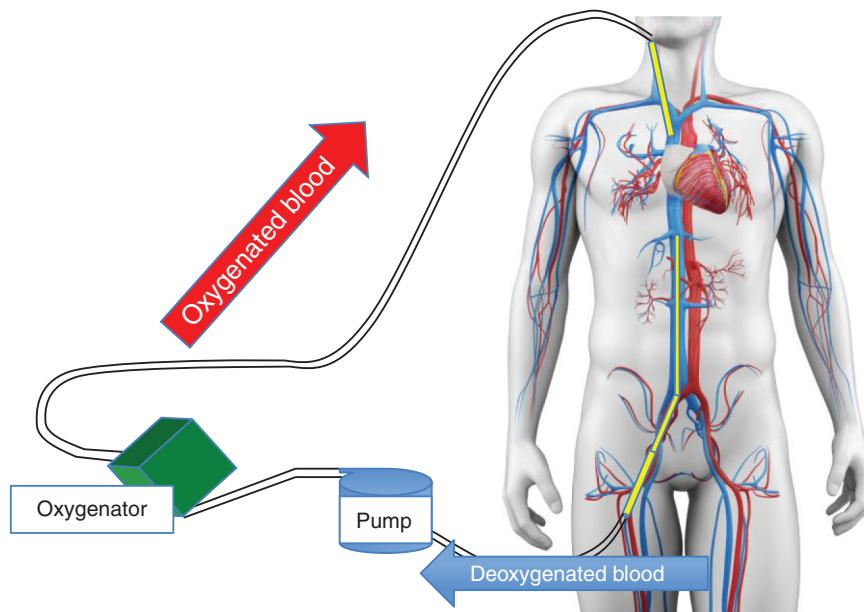


FIG. 8.1 The basic function of the ECMO circuit. (Modified from [SciePro/Shutterstock.com](#).)

Where this blood is drained from and where it returns to is exactly what we will go over in this chapter. We will cover a lot of configuration possibilities and when possible, the relevant physiologic rationale. This chapter is neither a comprehensive compilation, nor is every configuration mentioned used by every cannulating team, as some may be related to the preferences of the team. If anything, the breadth of this list underscores the many possibilities that exist for configuring the ECMO circuit to the specific physiologic needs of the patient.

CANNULA SELECTION

The first consideration is size and type of cannulas. Cannulas are large catheters that are inserted into heart or blood vessels by which blood is drained and returned. Most of the time, one cannula drains the blood while the other cannula returns the blood. These two cannulas often differ by where holes are placed for drainage and return of blood. There are primarily two types of cannulas that should be distinguished, referred to as multistage and single stage (Fig. 8.2).

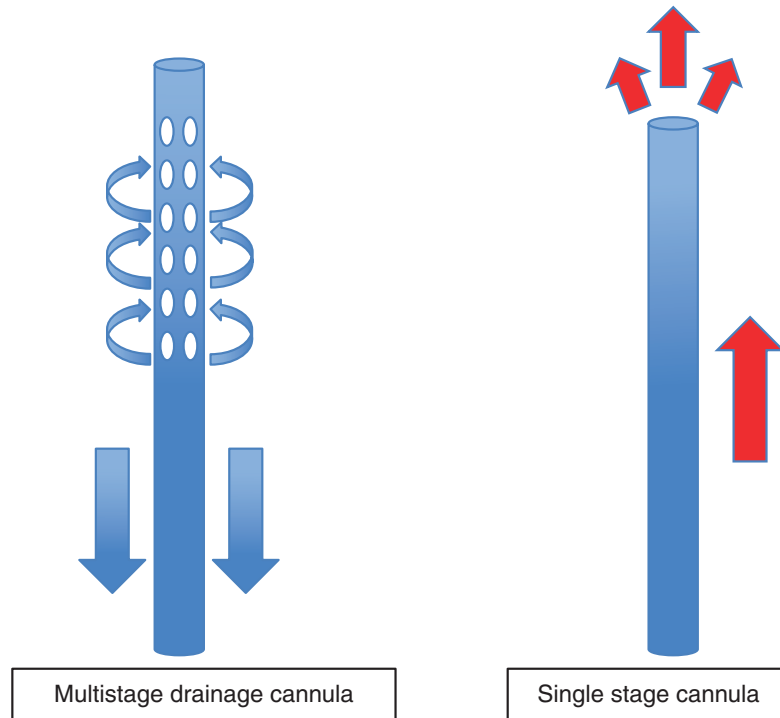


FIG. 8.2 Multistage drainage cannula (left) with multiple drainage holes and single stage cannula (right) with single point of exit for blood

Multistage cannulas are usually used as the drainage cannula. When draining blood, you want to have the ability to pull in blood over a large surface area so that a sufficient volume of blood can be run through the circuit without excessive negative pressure to do so. Thus, multistage cannulas have multiple holes for draining blood.

Compare this to the single-stage cannula, which has only one set of holes at the top. This carries the advantage of allowing for directional flow of returned blood. The single-stage return cannula can differ in length, usually being around 15 cm for short return cannulas and 50 cm for long return cannulas.

The other type of cannula is the dual lumen cannula. Dual lumen cannulas embed a return cannula inside an outer, drainage cannula as illustrated in Fig. 8.3. There is an outlet that allows for the return of oxygenated blood that can be aimed in a manner so that it does not get pulled back into the ECMO circuit. Dual lumen cannulas allow for both drainage and return of blood from a single access site, which can allow for improved mobility and ambulation.

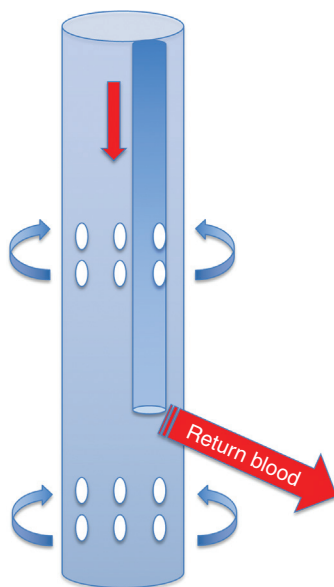


FIG. 8.3 Dual lumen cannula draining blood from outer lumen and returning blood through inner lumen

The final consideration for cannula selection is width and length. The length can vary but for the most part is fairly standard for multistage cannulas, all being around 50 cm for adults. The length of the return cannula depends on where the cannula is being implanted and for the most part remains as a long return (~50 cm) or a short return (~15 cm) (Fig. 8.4).

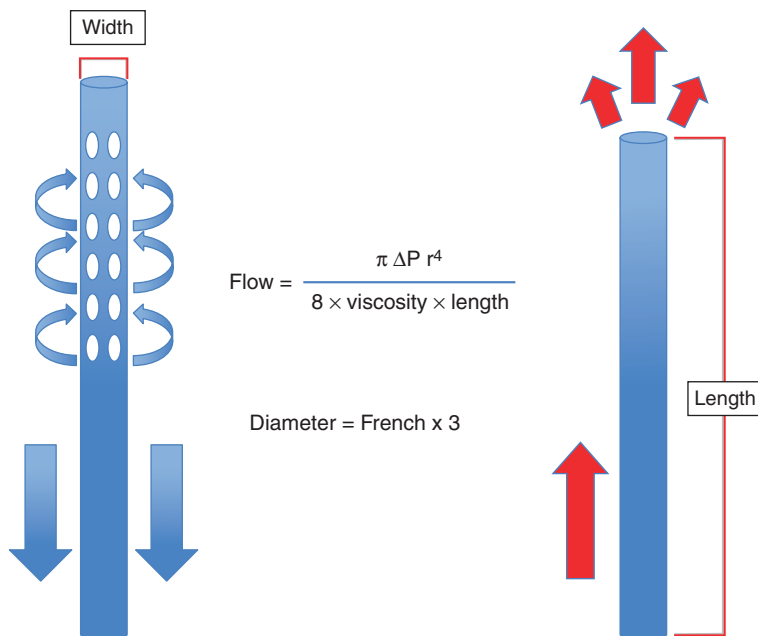


FIG. 8.4 Width and length considerations in cannula selection

Width is an important consideration for cannula selection, as the width is an important determination of overall flow – the wider the cannula, the more flow you will be able to obtain. We will talk about the concept of flow in much greater detail in [Chapter 10](#). The benefits of greater flow have to be weighed against the possibility of greater obstruction of the blood vessel, more damage to the blood vessels, and whether or not the vessels can accommodate a larger cannula.

The standard convention for width is French, which is three times the diameter of the cannula in millimeters. This can be helpful when sizing cannulas to the size of the blood vessel. If the blood vessel is 6 mm, for example, then it likely could not accommodate a 21-French cannula, which would have a diameter of 7 mm.

NAMING CONVENTION

First off, let's review the basic naming convention. Typically, the name of the configuration follows the direction of blood. If you are draining blood from the venous system and returning it to the arterial system, then this is referred to as **Veno-Arterial ECMO** or **VA ECMO** ([Fig. 8.5](#)).

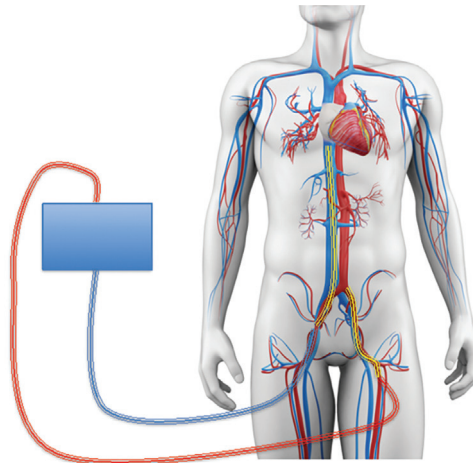


FIG. 8.5 Veno-arterial (VA) configuration. (Modified from [SciePro/Shutterstock.com](#).)

Contrast this to the following configuration, where we are draining from the venous system and returning back to the venous system. This is referred to as **Veno-Venous ECMO**, or **VV ECMO** ([Fig. 8.6](#)).

Now let's look at some configurations, starting with VV ECMO.

FEMORAL JUGULAR VV ECMO

Femoral-jugular involves a drainage cannula that is implanted into the inferior vena cava (IVC) via the common femoral vein and a short return cannula implanted into the superior vena cava (SVC)/right atrial junction via the internal jugular vein ([Fig. 8.7](#)). The drainage cannula is usually a multistage cannula to allow for optimal drainage and positioning. Ideally, the drainage cannula should be inserted into the intrahepatic IVC right under the diaphragm. The intrahepatic IVC position allows for the liver parenchyma to keep the IVC open and helps to optimize flow without collapse of the IVC around the negative pressure generated from the drainage ports.

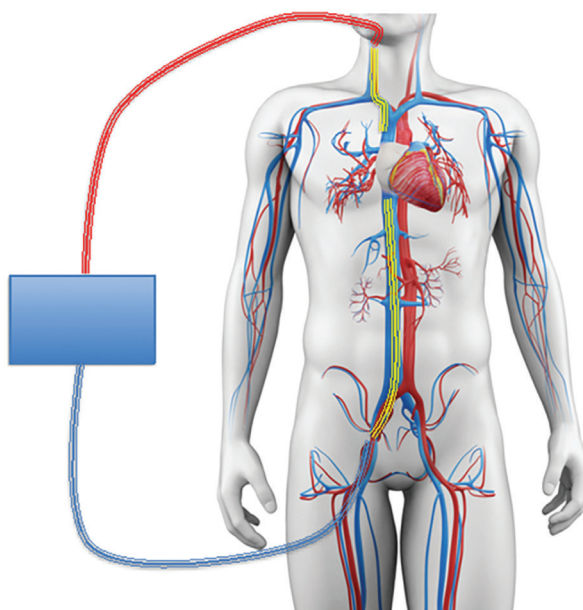


FIG. 8.6 Veno-venous (VV) configuration. (Modified from [SciePro/Shutterstock.com](#).)

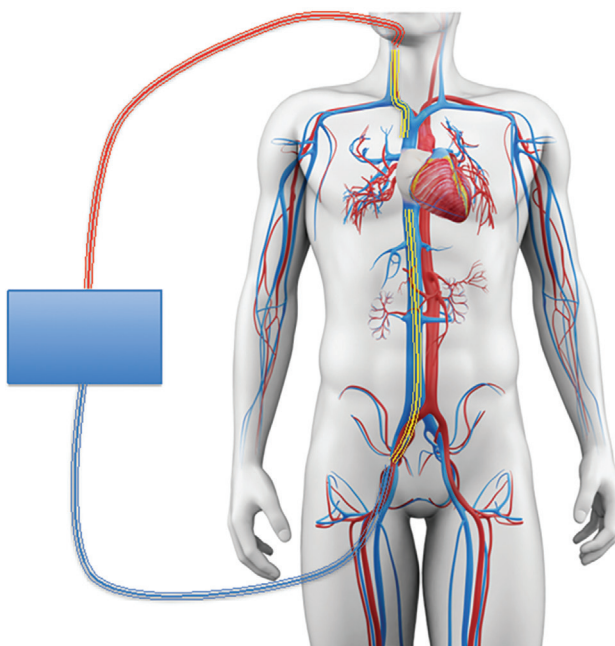


FIG. 8.7 Femoral jugular configuration. (Modified from [SciePro/Shutterstock.com](#).)

The return cannula, by contrast, should be a short, single-stage cannula, implanted into the right or left internal jugular vein, although the right IJ vein allows a somewhat straighter path to the optimal position in the RA/SVC junction. Placing the return cannula in the SVC/RA junction, with the drainage cannula lower at the intrahepatic IVC just below the diaphragm, separates the two

cannulas and prevents oxygenated blood that is being returned from the ECMO circuit to be sucked back into the circuit, a concept known as recirculation, which we will cover in more detail later on.

FEMORAL-FEMORAL VV ECMO

Femoral-femoral VV ECMO involves a multistage drainage cannula placed into the intrahepatic IVC just like in the femoral-IJ configuration; however, in this configuration, the return cannula is a long, single-stage cannula implanted into the IVC/RA junction from below, via that contralateral femoral vein (Fig. 8.8). Since the return cannula is a single-stage cannula, and is inserted above the top of the drainage cannula, blood can theoretically be returned without being sucked back into the drainage cannula in the form of recirculation. However, this risk of recirculation can be more significant with this configuration than with femoral-jugular.

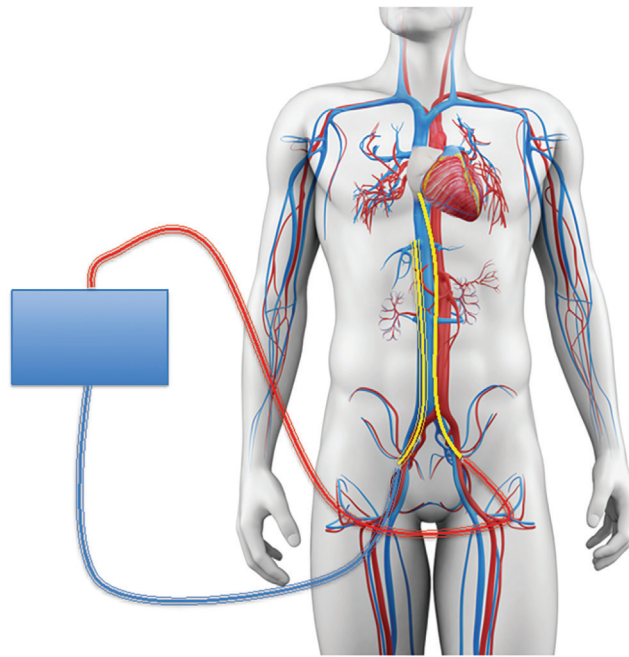


FIG. 8.8 Femoral-femoral configuration. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com/user/sciepro).)

ADDITIONAL VENOUS DRAIN

Intermittently, you may come across a circuit with two drainage cannulas. To confirm it is two drainage cannulas as opposed to two return cannulas, ensure that both cannulas are draining dark blood.

What is the advantage of an additional drainage cannula? Some providers use a second cannula if obtaining adequate flows are insufficient. Usually, if a second cannula is added for improved flow, it is a drainage cannula, as the flow limitations on VV ECMO are often dictated by the drainage side (Fig. 8.9).

This is a configuration that is rarely used, simply because a properly sized drainage cannula, placed into good position in the intrahepatic IVC, is often able to obtain more than adequate flows. However, there are rare instances (for example, in the development of abdominal compartment syndrome due to bleeding) where even a well-positioned drain cannula cannot obtain

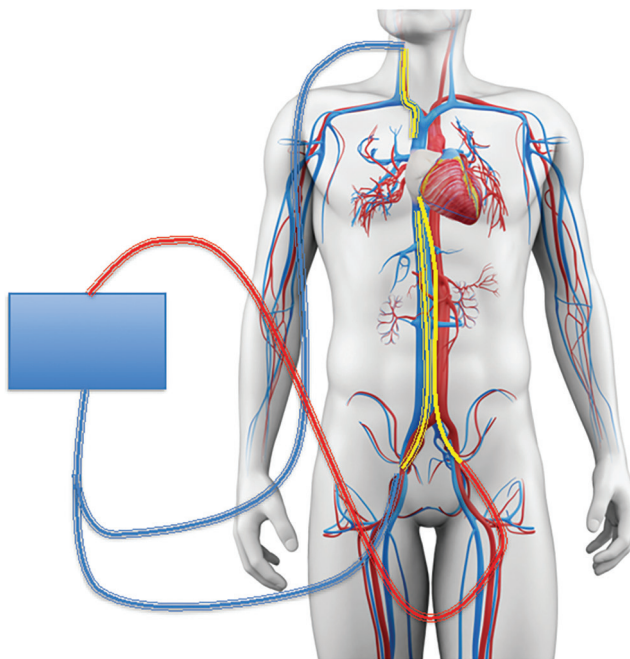


FIG. 8.9 Additional venous drain. (Modified from [SciePro/Shutterstock.com](#).)

the adequate flows. In this case, an additional drain cannula may be added and connected to the circuit through a Y-connection ([Fig. 8.23B](#)).

DUAL LUMEN CANNULA

The dual lumen cannula offers the ability to drain blood from the outer cannula with an inner cannula, which can return blood. In the standard cannula configuration, this is inserted in the intrahepatic IVC via the right internal jugular vein, which carries the benefit of being the straightest path. In the proper configuration, the outlet port lines up with the right atrium with the outflow track flowing across the tricuspid valve, with inlet ports positioned above and below the outlet ports ([Fig. 8.10](#)).

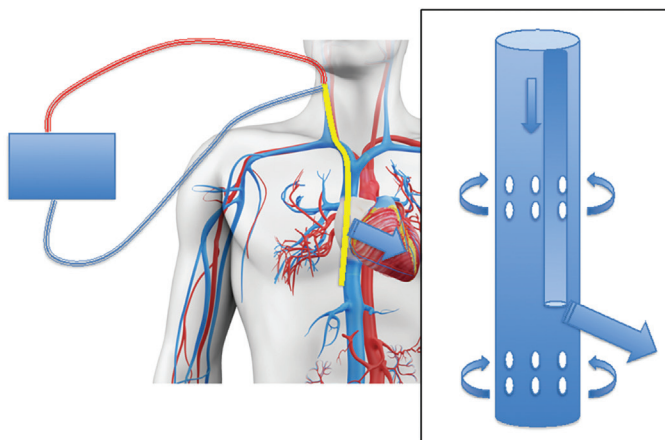


FIG. 8.10 Internal jugular vein dual lumen cannula. (Modified from [SciePro/Shutterstock.com](#).)

Dual Lumen Cannula in Subclavian Position

As compared to the IJ approach, the subclavian approach can be more stable, with less rotation and better positioning of the outflow track across the tricuspid valve. Consequently, this configuration can better facilitate mobility, with less worry about rotational positioning. However, this configuration can be more technically challenging to implant (Fig. 8.11).

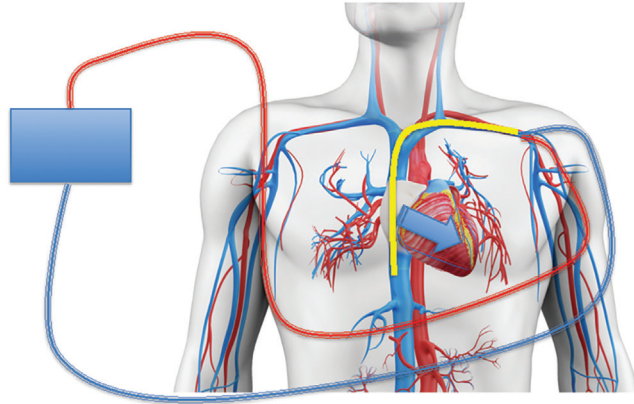


FIG. 8.11 Subclavian vein dual lumen cannula. (Modified from [SciePro/Shutterstock.com](#).)

Dual Lumen IJ-PA Configuration

A different dual lumen cannula offers the ability to drain from the SVC/right atrium and return blood directly into the pulmonary artery. This is accomplished by implanting the cannula from the IJ vein into the SVC/right atrium, across the tricuspid valve into the right ventricle, and across the pulmonic valve into the pulmonary artery, as illustrated in Fig. 8.12.

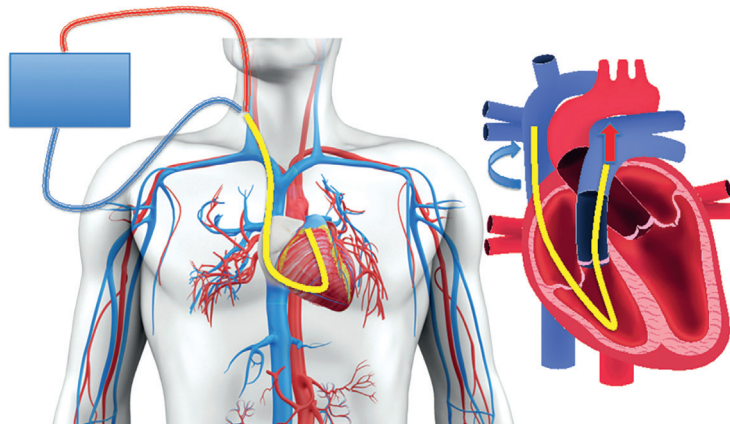


FIG. 8.12 Dual lumen IJ-PA cannula. (Modified from [SciePro/Shutterstock.com](#) and [Ody_Stocker/Shutterstock.com](#).)

This configuration requires a slightly different cannula, which is longer to reach all the way to the pulmonary artery that has the outflow track at the tip of the cannula instead of out the side.

This configuration carries the benefit of draining blood from the right ventricle and returning blood past the pulmonic valve, so it provides oxygenated blood in addition to providing some

right heart support. Additionally, there is theoretically less chance of recirculation as the pulmonary valve separates the inflow holes from the outflow track. However, this cannula can be more technically challenging to implant and manage.

FEMORAL-PA VV ECMO

If a dual lumen cannula to reach the pulmonary artery is not available or preferred, a similar configuration can be obtained by draining from the IVC via a multistage cannula implanted into the femoral vein and returning into the pulmonary artery via a long, single-stage, cannula implanted into the pulmonary artery via the IJ vein. This configuration would still provide VV ECMO support and right ventricular support but would require two-site cannulation (Fig. 8.13).

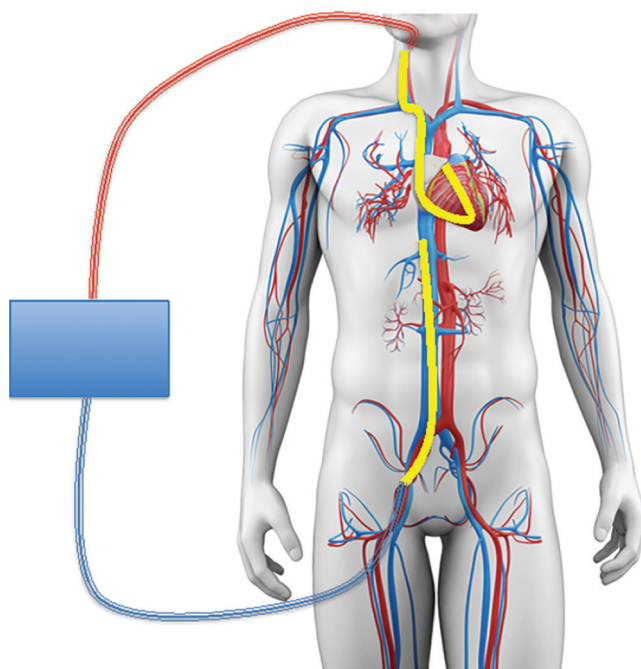


FIG. 8.13 Femoral-PA cannula. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com).)

VA ECMO: FEMORAL-FEMORAL

Percutaneously placed VA ECMO tends to be much more straightforward than VV ECMO. For adults, the near universal configuration is a multistage cannula implanted in the intrahepatic IVC via the femoral vein and a short, single-stage return cannula implanted into the distal aorta via the femoral artery (Fig. 8.14).

The arterial cannula directs blood retrograde towards the heart. Consequently, a short cannula works the best for this function, as a longer cannula would bypass much of the splanchnic and renal blood supply. We will go over the physiology related to this retrograde flow in much more detail in [Chapter 14](#).

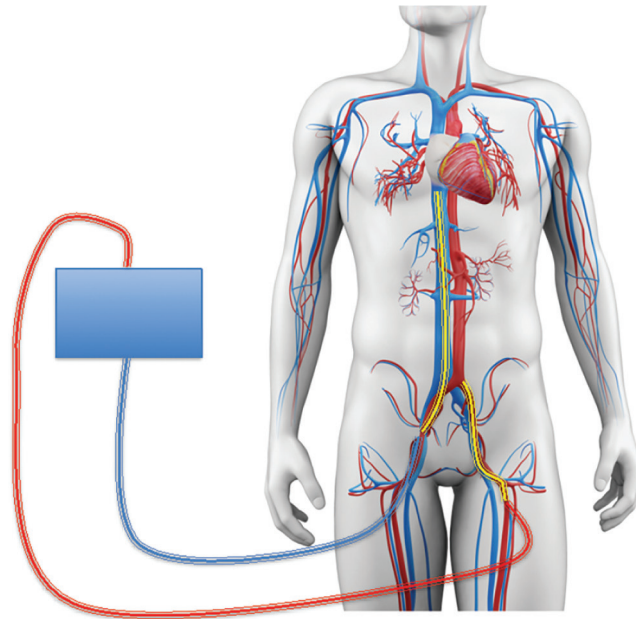


FIG. 8.14 Femoral-femoral VA ECMO configuration. (Modified from [SciePro/Shutterstock.com](#).)

ARTERIAL ACCESS FOR VA ECMO

Placement of an arterial cannula carries the unique challenge of the cannula impeding forward, arterial flow to the distal, ipsilateral extremity ([Fig. 8.15](#)).

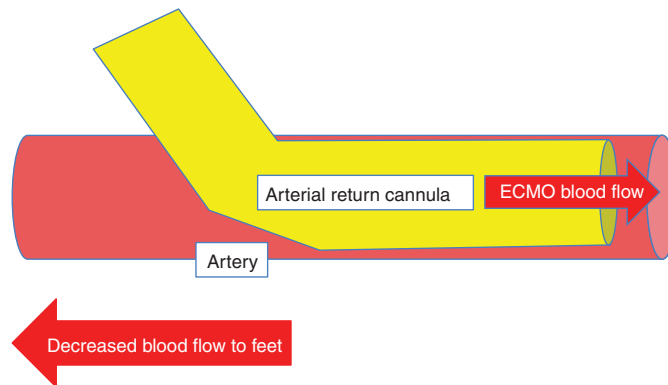


FIG. 8.15 Arterial cannula with obstruction of distal blood flow

There are a few strategies to mitigate this adverse effect:

1. Placement of a small enough cannula to allow for forward blood flow around the cannula
2. Placement of a graft on the artery to allow for flow both retrograde towards the aorta and antegrade down the leg
3. Placement of a distal perfusion cannula into the superficial femoral artery with diversion of oxygenated blood from the return line down the leg for better perfusion

Arterial grafts allow for blood flow in both directions but require a surgical cutdown and can have some technical considerations, as the angle of the graft can determine the amount of blood flow in either direction, potentially resulting in overflow/underflow (Fig. 8.16).

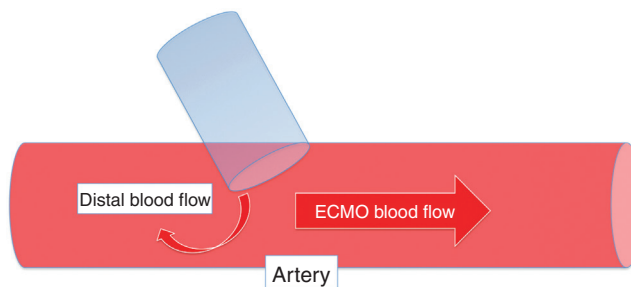


FIG. 8.16 Arterial graft with bidirectional blood flow

The distal perfusion catheter can potentially be placed percutaneously and can allow for blood flow directly from the ECMO circuit (Fig. 8.17).

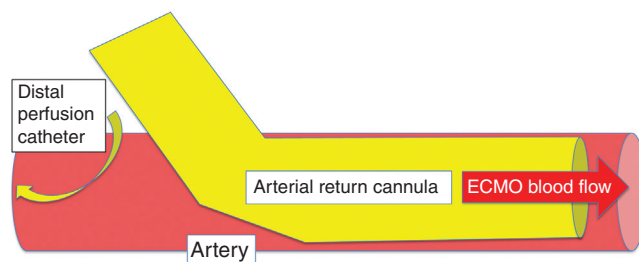


FIG. 8.17 Arterial cannula with distal perfusion catheter

The sheath can become occluded or kinked, so should be flushed intermittently (Fig. 8.18).

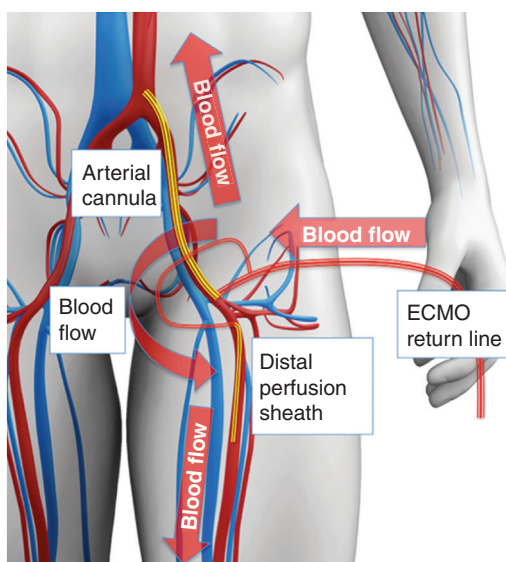


FIG. 8.18 Connection of distal perfusion catheter to return limb of the ECMO circuit
(Modified from SciePro/Shutterstock.com.)

VA ECMO: SURGICAL CANNULATION

Surgical access for VA ECMO allows for other configurations. The first is the standard graft that can be sewn on the artery, allowing for access to either the femoral artery or access to smaller vessels like the axillary artery (Fig. 8.19).

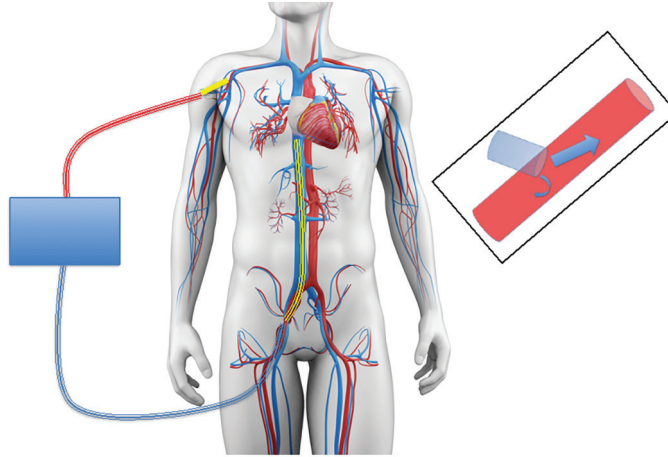


FIG. 8.19 VA ECMO with axillary graft. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com/user/sciepro).)

The axillary artery graft allows for more mobility as there is no limitation of a femoral arterial cannula. Additionally, it carries the added advantage of allowing for antegrade flow. There are technical challenges including the need to obtain the right angle, lest the arm gets too much or too little blood.

IJ AXILLARY CANNULATION

Other models can be employed to allow for complete upper body cannulation, for example, the IJ-axillary cannulation. This can further facilitate mobility (Fig. 8.20).

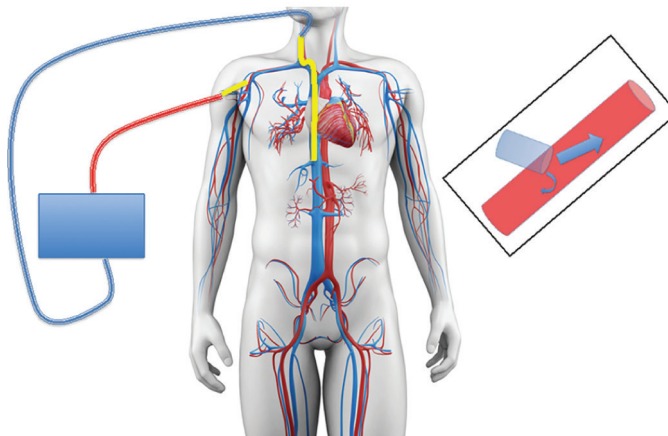


FIG. 8.20 IJ axillary cannulation. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com/user/sciepro).)

CENTRAL CANNULATION: FEMORAL AORTIC

If any cannulas/grfts are implanted directly into the heart or great vessels, this is often referred to as central cannulation. The return cannula is implanted into a graft sewn directly onto the aorta, which allows for antegrade flow. Since the chest is often open when cannulating in this manner, this is frequently combined with a drain in the left ventricle, which allows for the direct removal of blood, mitigating left ventricular distension in a process called venting. We will discuss this configuration in much more detail in [Chapter 15](#).

The drainage in these cases can either be a femoral drain, which is often implanted percutaneously, or a graft that is sewn directly onto the right atrium ([Figs. 8.21 and 8.22](#)).

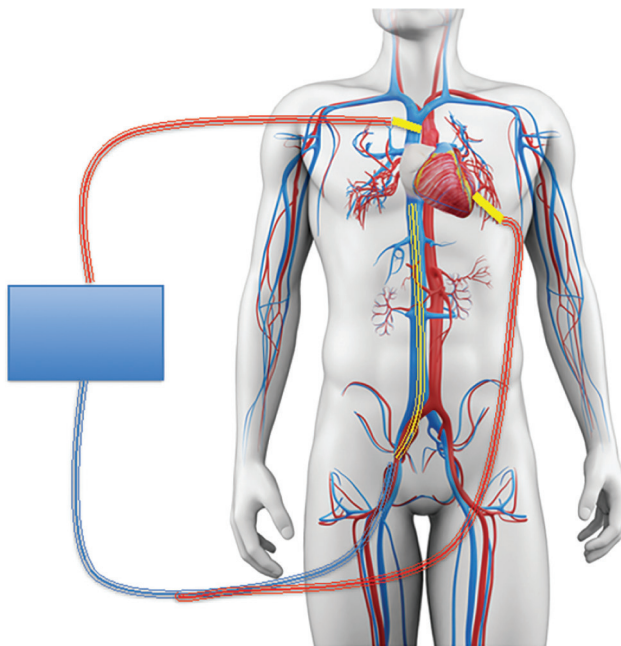


FIG. 8.21 Central cannulation: femoral-aortic. (Modified from [SciePro/Shutterstock.com](#).)

While there can be advantages to this cannulation strategy, often the chest is left open, which can have many ancillary effects to include risk of infection, inability to wean sedation/analgesic, immobility, and requirement of mechanical ventilation. In some cases, the cannulas can be tunneled out of the skin and the chest can be closed. However, there will exist the need for eventual return to the Operating room (OR) for removal.

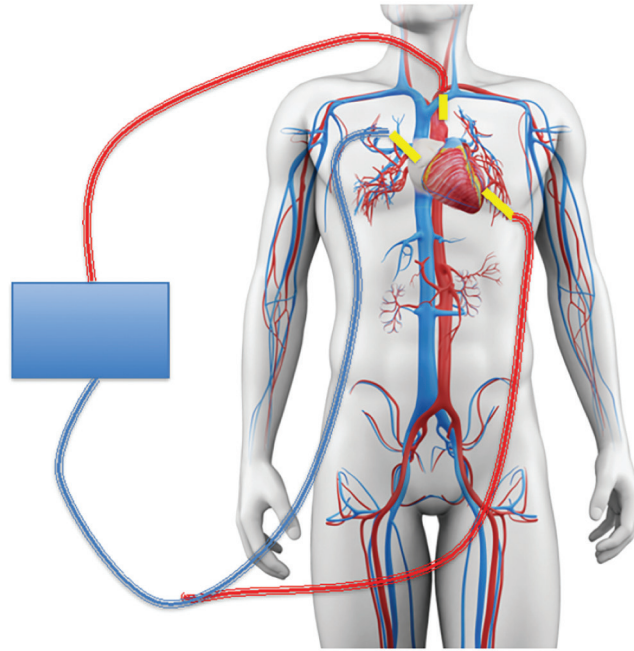


FIG. 8.22 Central cannulation: RA-aortic with left ventricular drain
(Modified from [SciePro/Shutterstock.com](#).)

HYBRID CONFIGURATIONS: WHERE VA AND VV MEET

Intermittently, additional cannulas can be added to a VV/VA circuit, in order to form a hybrid configuration. These cannulas drain blood that is connected through a Y connector that can direct the flow of blood in two directions as illustrated in [Fig. 8.23A,B](#).

This is where naming convention is important as it is easy to confuse how blood is flowing. Remember, the configuration is named in the direction of blood. If you are Y connecting two venous drains while maintaining one arterial return, then this is called VVA, as illustrated in [Fig. 8.24A,B](#).

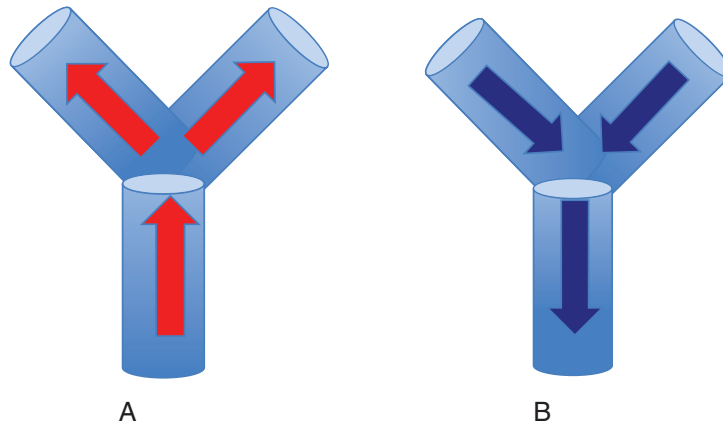


FIG. 8.23 A: Y connection for returned blood. B: Y connection for drained blood

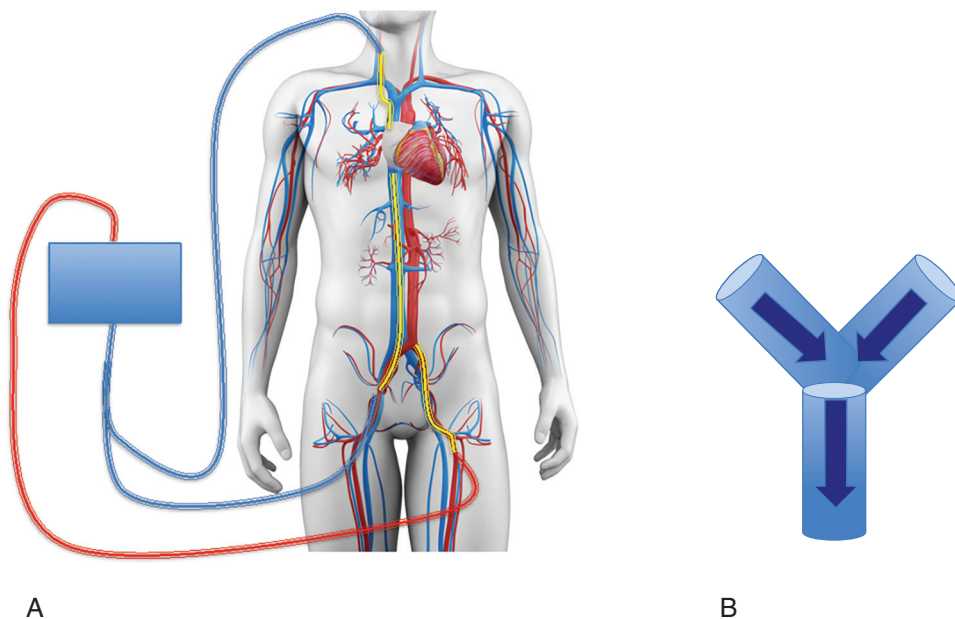


FIG. 8.24 VVA hybrid configuration with Y connection on the drainage side
(Modified from [SciePro/Shutterstock.com](#).)

Compare this to the more common hybrid configuration, where the return line is split between returning to an arterial and a venous cannula. This would be named as VAV ECMO, as illustrated in [Fig. 8.25A,B](#).

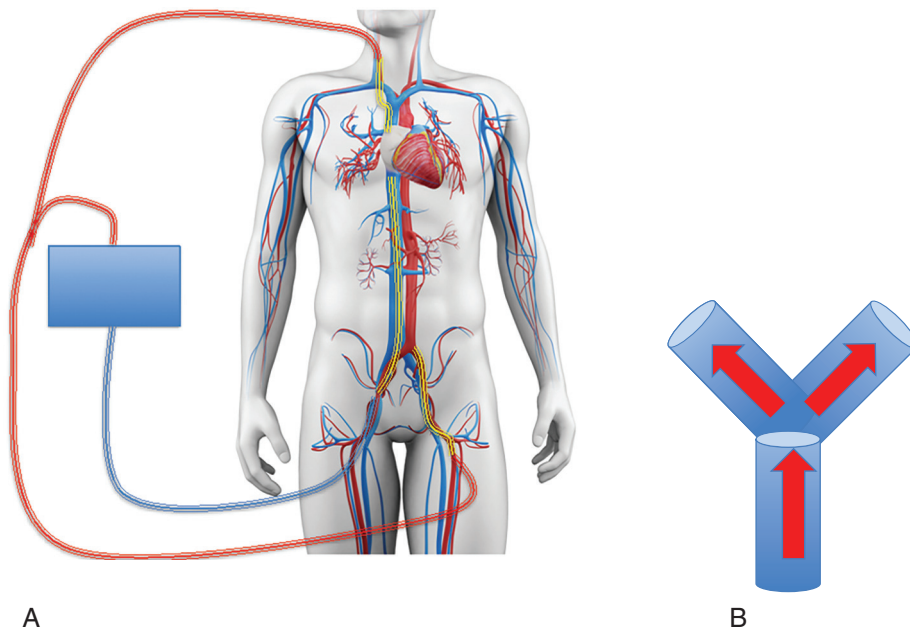


FIG. 8.25 VAV hybrid configuration with Y connection on the return side
(Modified from [SciePro/Shutterstock.com](#).)

This tends to be a more common configuration because there is a physiologic rationale for providing antegrade oxygenated blood in peripherally placed VA ECMO, where there may be the possibility of deoxygenated, competing flow that is being ejected by the heart. We will dive into this physiology in much more detail when we further explore retrograde physiology in [Chapter 14](#).

A FINAL WORD

We covered a lot of configurations in this chapter. Some you may commonly come across, while some can be rarely employed based on the physiologic result that is trying to be achieved. In either case, this emphasizes the importance of thoughtful consideration behind cannula selection and configuration. The team managing a patient on ECMO should have a prediction of what physiology they are likely to encounter and proceed accordingly with the appropriately sized cannulas and configuration.

Keep in mind, there is something to be said for simplicity and consistency. If you are able to achieve your anticipated physiologic endpoints, choosing cannula sizes and configurations that you are comfortable with understanding and managing can be prudent.

SUGGESTED READING

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